Full-production spectral Galerkin convergence: conditional theorem

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

first issued 2026-07-17 · this version issued 2026-07-17 · v2.1

1. Purpose and scope

This is the integrated referee package for the quantitative P3 theorem. The field is $u : \mathbb{T}^3 \to \mathbb{C}^3$, treated as six real components on the fixed torus with periods $16, 16, 16$. The canonical P1 functional has the pinned positive regularisers, $\eta_{\text{shell}} = 0$, and density floor $\varepsilon_\rho = 10^{-12}$. The P2 gradient flow is

$$\partial_t u + Lu + N(u) = 0. \tag{1.1}$$

The single named hypothesis is

$$\texttt{A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL}: \tag{H3}$$

the hash-pinned T5 P1 standalone reference functional is the canonical production functional. The T6 P2 well-posedness theorem is a hard dependency.

Theorem. Assume (H3), let $R \in \{0.5, 1, 2\}$, and let $\|u_0\|_{H^2} \le R$. For $\tau = 0.02$ and $T = 0.2$ the P2 solution satisfies

$$\sup_{\tau \le t \le T} \|u(t)\|_{H^6} \le B_6(R, \tau, T) < \infty. \tag{1.2}$$

For every even Fourier grid size N, with P_N the real-L^2 Fourier projector and R_N^C the pinned collocation residual,

$$\sup_{\tau \le t \le T} \|P_N R(u(t)) - R_N^C(P_N u(t))\|_{L^2} \le C(R, \tau, T) N^{-2}. \tag{1.3}$$

Let u_N instead solve the exact Galerkin equation

$$\partial_t u_N + L u_N + P_N N(u_N) = 0, \qquad u_N(\tau) = P_N u(\tau). \tag{1.4}$$

Then an explicit finite $E(R, \tau, T)$ satisfies

$$\sup_{\tau \le t \le T} \|u_N(t) - P_N u(t)\|_{L^2} \le E(R, \tau, T) N^{-4}. \tag{1.5}$$

Equations (1.2)–(1.5) are a positive-time, restarted, exact-Galerkin conditional theorem. They do not assert an algebraic rate from $t = 0$ for merely H^2 data and do not identify finite oversampling with exact Galerkin for the rational Class-II term.

1 Continuum and discrete conventions

The real pairing is

$$\langle v, w \rangle_{\mathbb{R}} = \operatorname{Re} \int_{\mathbb{T}^3} v^\dagger w,$$

and

$$\|v\|_{H^{2m}}^2 = |\mathbb{T}^3| \sum_k (1+|k|^4)^m |\widehat{v}_k|^2. \tag{2.1}$$

The exact Galerkin residual is $P_N R(v_N)$. The executable collocation residual R_N^C is the real gradient of the periodic collocation energy. Pointwise nonlinear products alias, so these objects are never conflated.

The linear symbol contains the Brazovskii fourth-order scalar part, family mass, and internal lock. Direct spectral bounds give

$$c_L \|v\|_{H^2}^2 \le \langle Lv, v\rangle \le C_L \|v\|_{H^2}^2, \tag{2.2}$$

where the derived values are $c_L = 0.2048572626782363$ and $C_L = 2.1566713536$.

2 Prior discretization evidence

The theorem is accompanied by six independently attackable evidence stages. The spatial residual and real-gradient audit passes 13/13. Its N8/N12/N16 collocation-to-projected-reference errors are $5.3301\,10^{-6}$, $4.2482\,10^{-10}$, and $6.6679\,10^{-12}$. The manufactured RK4 audit passes 10/10 with observed orders 4.016 and 4.008. The Hessian/Ritz audit passes 13/13, with homogeneous isolated-cluster gap 0.1368657 and maximum residual/gap about $1.01\,10^{-8}$.

An independently implemented portable functional gives an 11/11 continuum-quadrature-proxy pass after exact Fourier prolongation. The recorded CPU/CUDA complex128/64 matrix passes 6/6 on Torch 2.13.0+cu130 and one NVIDIA GeForce RTX 5070 Laptop GPU. These finite-grid results are supporting checks; the uniform theorem is established by the estimates below.

3 Explicit Fourier and nonlinear constants

The max-shell count $24m^2 + 2$, an integral tail, and Hausdorff–Young give the derived embeddings

$$\|v\|_\infty \le E_2 \|v\|_{H^2}, \quad E_2 = 0.329926816704, \qquad \|\partial_i v\|_{L^4} \le E_{14}\|v\|_{H^2}, \quad E_{14} = 0.533391254984. \tag{4.1}$$

The Fourier-weight inequality $w(k) \le 4\{w(\ell) + w(k-\ell)\}$ and Young convolution yield the deliberately enlarged algebra estimate

$$\|fg\|_{H^2} \le A_2 \|f\|_{H^2}\|g\|_{H^2}, \qquad A_2 = 8E_2 = 2.63941453363. \tag{4.2}$$

The Class-II ratios are $q_A = m_A/(\rho + \varepsilon_\rho)$. For $|u| \le A$, the inverse-density derivatives through order three obey

$$I_j = j!(2A+2)^j \varepsilon_\rho^{-(j+1)}, \qquad 0 \le j \le 3. \tag{4.3}$$

Leibniz convolution with the quadratic moment produces explicit Q_j for $D^j q_A$. A second convolution produces G_j for the Class-II metric. The Pauli-generator bounds $\|A_J\| \le 2|u|$ and $\|A_K\| \le 4|u|$ are enlarged by the declared component/spatial/generator contraction factor 108. Consequently the pinned nonlinear map has computed majorants

$$\|N(u)\|_{L^2} \le K_0(M_2), \quad \|N(u) - N(v)\|_{L^2} \le L_0(M_2)\|u - v\|_{H^2}, \tag{4.4}$$

and

$$\|N(u)\|_{H^2} \le K_2(M_2, M_4), \quad \|N(u) - N(v)\|_{H^2} \le L_2(M_2, M_4)\|u - v\|_{H^4}. \tag{4.5}$$

Every Q_j, G_j, K_s, L_s is persisted in the quantitative JSON; the small density floor is retained explicitly rather than hidden in an unnamed Moser constant.

4 Uniform H^2, H^4, and H^6 bounds

The energy-to-ball audit computes the all-time input

$$M_2(R)^2 = \frac{2\{E_+(R) + C_Y|\mathbb{T}^3|\}}{c_{H^2}}. \tag{5.1}$$

For $R = 0.5, 1, 2$, respectively, M_2 is bounded by 20.16099, 20.35978, and 21.21616.

For a positive self-adjoint L the spectral calculus gives

$$\|L^\alpha e^{-tL}\| \le (\alpha/(et))^\alpha. \tag{5.2}$$

Take $\alpha = 3/4$, $\theta = 1/4$, and $\delta = \tau/4$. The mild formula first bounds u in $D(L^{3/4})$. It also gives Holder continuity in $D(L^{1/2})$; (4.4) transfers that modulus to $N(u)$ in L^2. The endpoint cancellation identity

$$L\int_a^t S(t-s)f(s)\,ds = (I - S(t-a))f(t) + \int_a^t LS(t-s)\{f(s) - f(t)\}\,ds \tag{5.3}$$

changes the nonintegrable r^{-1} kernel to $r^{\theta-1}$ and gives an explicit M_4 on $[\tau/2, T]$. Repeating (5.2)–(5.3) on base space H^2 with (4.5) yields (1.2). The worst-case derived logarithmic bounds are

$$\log_{10} B_6 \le 515.378, qquad 515.751, qquad 517.320 \tag{5.4}$$

for the three declared radii. These immense but finite values are evaluated with decimal arithmetic.

5 Residual and aliasing theorem

The omitted Fourier cube satisfies

$$\|(I - P_N)v\|_{H^2} \le (L_{\max}/\pi)^4 N^{-4}\|v\|_{H^6}. \tag{6.1}$$

For periodic sampling, Cauchy–Schwarz over alias shells and

$$\sum_{m\ge1} \frac{24m^2+2}{m^4} = 4\pi^2 + \frac{\pi^4}{45}$$

give

$$\|P_N(I_N - I)f\|_{L^2} \le A_{\text{alias}} N^{-2}\|f\|_{H^2}, \qquad A_{\text{alias}} = 167.383081344. \tag{6.2}$$

Substitution of (4.4)–(5.4) into (6.1)–(6.2) proves (1.3). For $R = 0.5, 1, 2$, the conservative derived enclosures are

$$\log_{10} C \le 576.355, qquad 576.770, qquad 578.513. \tag{6.3}$$

6 Exact-Galerkin finite-time evolution

The finite-dimensional restriction (1.4) preserves the gradient-flow energy identity and the same H^2 envelope. For $e = u_N - P_N u$, coercivity, (4.4), and (6.1) yield

$$\frac{d}{dt}\|e\|_{L^2} \le a_R\|e\|_{L^2} + L_0(M_2)(L_{\max}/\pi)^4 B_6 N^{-4}, \qquad a_R = \frac{L_0(M_2)^2}{2c_L}. \tag{7.1}$$

Since $e(\tau) = 0$, Gronwall proves (1.5) with

$$E = L_0(M_2)(L_{\max}/\pi)^4 B_6 \frac{e^{a_R(T-\tau)} - 1}{a_R}. \tag{7.2}$$

The JSON stores $\log_{10} E$, since the deliberately pessimistic floor-based exponential is beyond ordinary floating-point range. Thus (1.5) is a genuine convergence theorem but not a useful N32/N64/N128 error bar.

7 Independent checks and reproduction

The primary quantitative audit passes 15/15. An independent implementation does not import the primary code: it reconstructs the linear spectrum and semigroup envelope, checks the N^{-4} projection tail and alias identity, probes exact directional derivatives of q_A against (4.3), and recomputes the decimal logarithms. It passes 10/10. The hash-bound integrated verifier checks all prior stages, the two quantitative audits, and recorded CUDA evidence at 104/104.

The one-command reproduction is

```
python codes/foundations/a3_full_production_verify.py --reuse-recorded-audits
```

and must print `ASSERTS: 104/104` followed by `A3-FULL-PRODUCTION-VERIFY-PASS`.

8 Devil's-advocate review

1. **Finite constants imply useful practical error bars. UPHELD as false.** The 10^{-12} floor makes the bounds astronomically large. The theorem establishes convergence only.
2. **H^2 data have the stated rate from $t = 0$. UPHELD as false.** Smoothing is used only for $t \geq \tau > 0$, and the Galerkin comparison is restarted at τ.
3. **A three-halves rule is exact for the Class-II residual. UPHELD as false.** The density denominator is rational. Only the mathematical projection $P_N N(u_N)$ is called exact Galerkin.
4. **The floor can be dropped because $|q_A| \leq 1$. VALID WITH MITIGATION.** Derivatives of q_A control the Moser and Lipschitz constants, so (4.3) retains the full floor dependence.
5. **The numeric constants could be pasted or fitted. DISMISSED.** Production inputs are source-hash pinned; all displayed majorants are labelled DERIVED and recomputed from those inputs.
6. **The finite-grid matrix proves the theorem. UPHELD as false.** Stages 1–5 are adversarial checks. The uniform statement rests on the analytic estimates (4.1)–(7.2).
7. **T6 silently inherits a T5 functional. VALID WITH MITIGATION.** The exact identification is promoted to the named hypothesis (H3), satisfying dependency monotonicity.

9 Falsifier and no-overclaim boundary

The theorem is falsified by source-hash drift; a failed 104-assertion verifier; an incorrect Fourier shell, semigroup, endpoint-cancellation, coefficient, or alias estimate; a counterexample to the stated positive-time bound; or failure of the exact Galerkin energy identity. Such a failure reopens `A3-FULL-DISCRETIZATION-CLOSURE`.

This theorem does not cover $t = 0$ algebraic rates for merely H^2 data, $\eta_{\text{shell}} \neq 0$, removal of either regulariser, infinite volume, fixed finite-oversampling exactness, a computably useful error at practical grids, historical Sector-B trajectories, a full-spectrum Hessian theorem, minimizer or BCC selection, T7, or physical-domain closure.

Result footer

```
Result ID:              A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:      Explicit B6 and residual C on [tau,T], plus N^-4
                        convergence of exact Galerkin restarted at tau.
```

Scope:	T6 CONDITIONAL-THEOREM; fixed T3, canonical P1/P2, eta_shell=0, R=0.5/1/2, tau=0.02, T=0.2.
Named hypothesis:	A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL.
Evidence grade:	ANALYTIC + EXECUTED + CONDITIONAL.
Reproduction command:	python codes/foundations/ a3_full_production_verify.py --reuse-recorded-audits
Expected output:	ASSERTS: 104/104; A3-FULL-PRODUCTION-VERIFY-PASS.
Falsification gate:	Failed source hash, coefficient/endpoint/Fourier estimate, exact-Galerkin identity, or verifier.
Tier before / after:	T4 / T6, one-shot conditional theorem justified by full proof, named hypothesis, independent audit, quantitative sanity checks, and published bundle.
No-overclaim statement:	Not a practical solver error bar, t=0 H2 rate, finite-oversampling exactness, Sector-B certificate, T7, minimizer, or BCC result.
Next required action:	Preserve as the closed P3 baseline; any sharper constants or executable solver bridge is separate.